Exercise careful thought and your problem solving skills (types of variables, variable names, choice of constructs, algorithmic steps) before you begin writing each program.

1. Write a program to print the day of the week, depending upon the number entered (e.g. 1 - Sunday, 2 - Monday, … 7 - Saturday).
   a. Use if statements
   b. Use switch

2. Write a program that accepts an integer (associated with a month e.g. 1 - January, 2 - February, … , 12 - December), and prints the number of days in that month.
   a. Use if statements
   b. Use switch

   Note: Your program should perform error checking e.g. numbers not within the range of 1 to 12.

3. Write a program to accept a letter and determines whether or not it is a vowel.

4. Using a switch statement determine the maximum of two numbers.

5. Write an program to print 1/10, 1/100, 1/1000, 1/10000, 1/100000.

6. Re-write your program in the previous question, by using a loop.

7. Write a program that prints integers from 1 to 100.
   a. Use a for loop
   b. Use a while loop
   c. Use a do..while loop

8. Write a program that prints integers from 100 to 1 (reverse order).

9. Write a program that prints even numbers between 1 and 100.

10. Write a program that prints odd numbers between 1 and 100.

11. Write a program that adds the numbers from 1 to 100, and output the result.

12. Using a for loop, count from 1 to 100 by 3s.

13. Using a while loop, count from 1 to 100 by 3s.

14. Using a do...while loop, count from 1 to 100 by 3s.

15. Write a program that reads 2 numbers between 1 and 100. The program should then add all numbers included within the range of numbers that were entered and print the result. For example, if the numbers 21 and 60 were entered, then the numbers 21, 22, 23 … 59 and 60 should be added.

16. A program that allows the user to convert measurements from ***either*** feet to meters, or minutes to hours, ***or*** degrees Fahrenheit to degrees Celsius. Design the program so that it displays a menu of options and allow the user to select one of the options, depending on what conversion he/she requires. The available options are:
    1. Display this menu
    2. Convert minutes to hours
    3. Convert feet to meters
    5. Quit.

    Use the following conversion table:

    1 foot = 0.3048 meter
    60 minutes — 1 hour

    HINT: Use loops and if or switch statements.

17. Write a program to determine the factors of an integer.

18. Write a program to accept an integer value and then print it in reverse.

19. Write a program to determine whether or not an integer is a palindrome.

20. Write a program to determine if an integer value is a prime number.